

Name: J. Vishnu Vardhan
Reg No: 192324297

Annexure A

Pseudocode Writing Task (Algorithm Design)

1. Face Detection and Recognition Using Eigenfaces

Scenario

Design an algorithm to detect multiple faces in an image and recognize them using the Eigenfaces method.

Input

Grayscale image of size $M \times N$

Output

Bounding boxes of detected faces and corresponding recognized names.

Algorithm: FACE_DETECTION_AND_RECOGNITION

ALGORITHM FACE_DETECTION_AND_RECOGNITION(Image, FaceDatabase)

INPUT:

- Image
- FaceDatabase containing known face Eigenface features

OUTPUT:

- Bounding boxes of detected faces
- Corresponding recognized names

BEGIN

1. Read the input Image
2. Convert Image into grayscale
IF the image is already grayscale
Keep the image unchanged

END IF

3. Normalize the grayscale image

Normalize pixel values to a standard range

4. Apply Histogram Equalization

This improves contrast and handles varying lighting conditions

5. Detect faces using Viola-Jones Face Detector

6. Store all detected face regions in FaceList

7. FOR each FaceRegion in FaceList DO

a. Extract the bounding box of FaceRegion

b. Crop the face using the bounding box

c. Resize the cropped face to the standard training size

d. Normalize the face image

e. Subtract the mean face from the input face

f. Project the face onto the Eigenface space

FaceFeature = EigenfaceProjection(FaceImage)

g. Set MinimumDistance = infinity

Set RecognizedName = "Unknown"

h. FOR each KnownFace in FaceDatabase DO

Calculate EuclideanDistance between
FaceFeature and KnownFace.Feature

IF EuclideanDistance < MinimumDistance THEN

MinimumDistance = EuclideanDistance

RecognizedName = KnownFace.Name

END IF

END FOR

i. IF MinimumDistance > RecognitionThreshold THEN

RecognizedName = "Unknown"

END IF

j. Draw the bounding box around the detected face

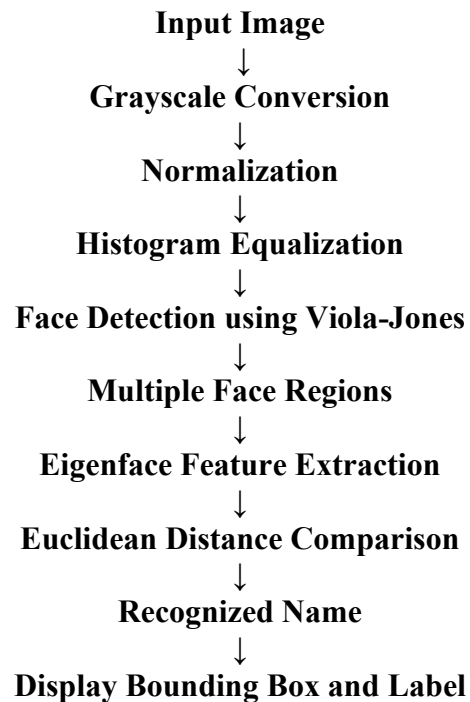
k. Display or store the RecognizedName
near the bounding box

END FOR

8. Return all bounding boxes and recognized names

END

Overall Flow



2. Real-Time Road Lane Detection in Driving Video

Scenario

Design an algorithm to detect road lanes from a real-time front-facing driving camera.

Input

Video frames from a front-facing camera.

Output

Lane lines highlighted on every frame.

Requirements

- Robust to shadows
- Detect curved roads
- Handle partial occlusion
- Process frames in real time

Algorithm: REAL_TIME_LANE_DETECTION

ALGORITHM REAL_TIME_LANE_DETECTION(Video)

INPUT:

Video frames from front-facing camera

OUTPUT:

Video frames with detected lane lines highlighted

BEGIN

1. Open the input Video
2. WHILE Video has frames DO
 - a. Capture the next Frame
 - b. Convert Frame to grayscale
 - c. Apply Gaussian Blur
to reduce image noise
 - d. Apply Histogram Equalization
to improve visibility under shadows
 - e. Define the Region of Interest (ROI)
Select only the road area
 - f. Apply Canny Edge Detection
to detect lane boundaries
 - g. Apply Perspective Transformation
IF curved lane detection is required
 - h. Apply Hough Transform
to detect lane line segments
 - i. Filter detected lines based on:
 - Lane position
 - Line slope

- Line length
- Continuity

j. Combine line segments to form:

- Left lane
- Right lane

k. IF curved lanes are detected THEN
 Fit a polynomial curve
 to the detected lane points
END IF

l. IF a lane is partially occluded THEN
 Use lane information from previous frames
 to estimate the missing lane section
END IF

m. Draw the detected left and right lanes
on a blank lane image

n. Overlay the lane image
on the original Frame

o. Display the final Frame

END WHILE

3. Close the Video

END

Simplified Pseudocode

BEGIN

Open Video

WHILE video is running DO

 Capture Frame

 Convert Frame to Grayscale

 Apply Gaussian Blur

 Apply Histogram Equalization

```
Select Region of Interest

Apply Canny Edge Detection

Apply Hough Transform

Detect Left and Right Lane Lines

IF road is curved THEN
    Apply Curve Fitting
END IF

IF lane is partially occluded THEN
    Use Previous Frame Information
END IF

Draw Lane Lines

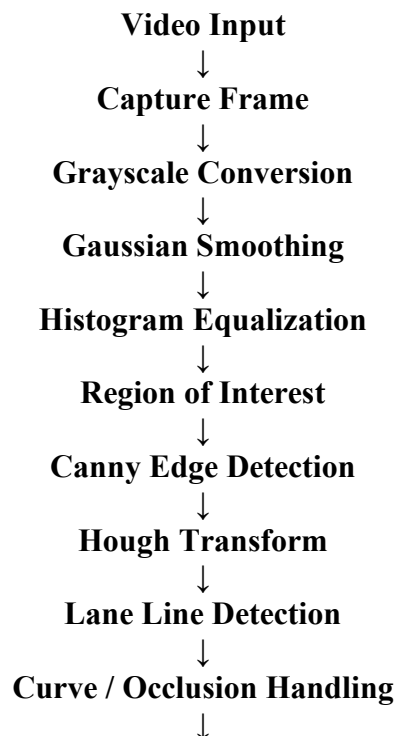
Overlay Lanes on Original Frame

Display Output Frame

END WHILE

END
```

Overall Flow



Overlay Lanes on Original Frame



Display Frame



Repeat for All Frames

Conclusion

The first algorithm detects multiple faces, preprocesses them to handle lighting variations, extracts Eigenface features, and recognizes faces using Euclidean distance. The second algorithm processes each video frame continuously to detect and highlight road lanes while handling shadows, curves, and partial occlusion.